

CT 2 Lab Component (Units 3 and 4)

Q1. Feature Engineering with Spark MLlib

Use Case: Preprocessing e-commerce transaction data for fraud detection.

Procedure:

1. Load a sample dataset (CSV of transactions) into Spark DataFrame.
2. Apply preprocessing: handle missing values, normalize numerical features, encode categorical variables.
3. Use Spark MLlib's **VectorAssembler** to combine features into a single vector.
4. Save the transformed dataset for model training.

Deliverables:

- Preprocessed dataset in Spark DataFrame.
- Feature vector file ready for MLlib models.

Q2. Classification Model with Spark MLlib

Use Case: Predicting customer churn in a telecom dataset.

Procedure:

1. Load telecom customer dataset into Spark.
2. Split into training and test sets.
3. Train a classification model (e.g., Logistic Regression or Decision Tree) using Spark MLlib.
4. Evaluate accuracy, precision, recall, and F1 score.

Deliverables:

- Trained classification model.
- Performance metrics report.

◆ Moderate Level

Q3. Regression Model with Spark MLlib

Use Case: Predicting house prices using real estate dataset.

Procedure:

1. Load housing dataset (features: area, bedrooms, location index, etc.) into Spark.

2. Apply feature engineering (scaling, encoding).
3. Train a regression model (Linear Regression or Random Forest Regressor).
4. Evaluate using RMSE and R^2 metrics.

Deliverables:

- Regression model file.
- Prediction results with error metrics.

Q4. Stream Processing with Kafka + Spark Streaming

Use Case: Real-time monitoring of IoT sensor data (temperature readings).

Procedure:

1. Set up Kafka producer to stream sensor data.
2. Configure Spark Streaming to consume Kafka topics.
3. Apply transformations (filter abnormal readings, compute moving averages).
4. Output alerts for anomalies into another Kafka topic or dashboard.

Deliverables:

- Kafka topic with processed stream.
- Real-time anomaly detection logs.

Q5. Workflow Orchestration with Airflow

Use Case: Automating ETL pipeline for daily sales data.

Procedure:

1. Define Airflow DAG with tasks:
 - Extract sales data from source (CSV/DB).
 - Transform data (cleaning, aggregation in Spark).
 - Load into data warehouse (Hive/BigQuery).
2. Schedule DAG to run daily at midnight.
3. Monitor execution via Airflow UI.

Deliverables:

- Airflow DAG script.
- Execution logs showing successful ETL runs.